

Jorge Garcia

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EDUCATION

University of Texas-Austin Austin, Texas
Computer Science (B.S.), GPA: 3.5946

August 2022 – May 2026

SKILLS

Soft: Communication, multitasking, leadership, organizational, goal oriented, detail-oriented, problem solving, time management, teamwork, adaptability

Programming: JavaScript, HTML, CSS, Python, Java, C, Linux, Bash, MATLAB, PowerShell, Go, Zig, Rust, C++, Vlang

Computer: Adobe Photoshop certified, Adobe Illustrator certified, Microsoft Word Expert certified

Languages: Native Spanish, Native English, Beginner French

Relevant coursework: Data Structures and Algorithms, Discrete Mathematics, Computer Architecture and Organization, Principles of Computer Systems, Algorithms, Compilers, Neural Networks, Advanced Computer Architecture, Database Management Systems

RESEARCH

RTX, University of Texas – Austin

June 2024 – August 2024

Researcher (Mentor: Dr. Krista Soderlund)

- Used MATLAB in conjunction with PowerShell scripts to calculate pressure of Europa's ice shell using varying inputs and created maps demonstrating various parameters.
- Analyzed maps and graphs as a function of two specific inputs and drew conclusions.
- Presented conclusion, figures, and other findings in a poster format to a research symposium.

RTX, University of Texas – Austin

June 2023 – August 2023

Researcher (Mentor: Dr. Thorston Becker)

- Wrote scripts using Bash and Python alongside a C library to calculate data from earthquakes and create geologic maps based on given numerical outputs.
- Interpreted maps through a space and time perspective and deduced results.
- Presented results, maps, and other information in a poster format to a research symposium.

PROJECTS

AruXI Project (Custom Architecture & Toolchain)

- Designed a custom 32-bit RISC instruction set architecture (Aru32) with arithmetic, branching, data movement, and privileged instructions.
- Implemented an emulator capable of running programs under the ISA, including condition flags and kernel/user privilege modes.
- Built an assembler and linker supporting the ISA with relocatable binaries and library support.

Command Line Dungeon-Crawler

- Designed and developed a terminal-based dungeon crawler game in C.
- Implemented game mechanics such as combat, inventory, items, and level generation

Mini OS

- Expanded a toy OS to include priority scheduling, stack-based argument passing, and a multi-threaded using a multi-level indexed scheme.
- Designed and implemented system calls for user programs and multi-level paged virtual memory.

Computer Architecture Animations

- Created animations using the MANIM python library to visualize concepts in computer architecture and organization.
- Demonstrated endianness, unsigned and signed integer addition and subtraction, memory padding for structs and unions, basic instruction pipeline for ARM.